Day 2 – Advanced Conditional Logic & Logical Functions

Module 1: Advanced Conditional Logic

What are Advanced Conditional Functions?

Advanced Conditional Functions allow us to evaluate **multiple conditions** instead of just one. They are commonly used in dashboards, reports, and business decision-making.

Why Do We Need Advanced Conditional Logic?

The basic IF function can evaluate only one condition at a time.

Example:

=IF(E2>=5,"Bulk","Normal")

But what if we want to classify Quantity into:

- High
- Medium
- Low
- Very Low

Using multiple nested IF statements becomes difficult to read and maintain.

Excel provides better alternatives such as:

- IFS
- AND
- OR

Difference Between IF and IFS

IF	IFS
Tests one condition	Tests multiple conditions
Uses Nested IF for multiple conditions	No Nested IF required
Difficult to read when conditions increase	Easy to read and maintain

Better for simple decisions	Better for multiple business rules

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IFS Function

Definition

The **IFS** function evaluates multiple conditions in sequence and returns the value corresponding to the **first TRUE condition**.

Unlike Nested IF, it does not require multiple IF functions inside one another.

Syntax

=IFS(logical_test1,value_if_true1,

logical_test2,value_if_true2,

...)

Syntax Explanation

Argument	Meaning
logical_test1	First condition to evaluate
value_if_true1	Result if first condition is TRUE
logical_test2	Second condition
value_if_true2	Result if second condition is TRUE
...	Continue adding conditions

Business Example

Classify Quantity Sold

Quantity	Category
8 or more	High
5–7	Medium
2–4	Low
Less than 2	Very Low

Formula

=IFS(

E2>=8,"High",

E2>=5,"Medium",

E2>=2,"Low",

E2<2,"Very Low")

Business Use Cases

- Employee Performance Rating
- Customer Segmentation
- Sales Performance
- Student Grade Classification
- Risk Analysis

Advantages of IFS

- Cleaner formulas
- Easier debugging
- Better readability
- No deep nesting
- Easier to modify

Module 2: Advanced Conditional Aggregate Functions

SUMIFS Function

Definition

The **SUMIFS** function adds values only when **multiple conditions** are satisfied.

Unlike SUMIF, SUMIFS supports two or more criteria.

Syntax

=SUMIFS(sum_range,

criteria_range1,criteria1,

criteria_range2,criteria2,...)

Syntax Explanation

Argument	Meaning
sum_range	Cells to be added
criteria_range1	First range to check
criteria1	First condition
criteria_range2	Second range
criteria2	Second condition

Example

Total Quantity Sold in Delhi for Laptops

=SUMIFS(E2:E1002,

C2:C1002,"Delhi",

D2:D1002,"Laptop")

Business Uses

- Sales by Product & City
- Revenue by Employee & Month
- Expenses by Department & Category

COUNTIFS Function

Definition

Counts records satisfying multiple conditions.

Syntax

=COUNTIFS(criteria_range1,criteria1,

criteria_range2,criteria2,...)

Syntax Explanation

Argument	Meaning
criteria_range1	First range
criteria1	First condition
criteria_range2	Second range
criteria2	Second condition

Example

Count Laptop Orders in Mumbai

=COUNTIFS(

C2:C1002,"Mumbai",

D2:D1002,"Laptop")

Returns

Number of matching records.

AVERAGEIFS Function

Definition

Calculates the average of values meeting multiple conditions.

Syntax

=AVERAGEIFS(

average_range,

criteria_range1,criteria1,

criteria_range2,criteria2)

Syntax Explanation

Argument	Meaning
average_range	Values to average
criteria_range1	First range
criteria1	First condition
criteria_range2	Second range
criteria2	Second condition

Example

Average Laptop Price in Delhi

=AVERAGEIFS(

F2:F1002,

C2:C1002,"Delhi",

D2:D1002,"Laptop")

Difference Between IF, IFS and *IFS Functions

Function	Conditions	Purpose
IF	One	Decision Making
IFS	Multiple	Multiple Decisions
SUMIFS	Multiple	Conditional Sum
COUNTIFS	Multiple	Conditional Count
AVERAGEIFS	Multiple	Conditional Average

Module 3: Logical Functions (AND & OR)

What are Logical Functions?

Logical Functions evaluate one or more conditions and return either:

- TRUE
- FALSE

These functions are commonly combined with IF to create powerful business rules.

AND Function

Definition

The **AND** function returns **TRUE only if every condition is TRUE**.

If even one condition is FALSE, the result is FALSE.

Syntax

=AND(logical1,logical2,...)

Syntax Explanation

Argument	Meaning
logical1	First condition
logical2	Second condition
...	Additional conditions

Example

Check if:

Quantity ≥ 5

AND

Price ≥ 10000

=AND(E2>=5,F2>=10000)

Returns

TRUE

Only when both conditions are TRUE.

Truth Table

Condition 1	Condition 2	Result
TRUE	TRUE	TRUE
TRUE	FALSE	FALSE
FALSE	TRUE	FALSE
FALSE	FALSE	FALSE

Business Uses

- Employee Eligible for Bonus
- Customer Eligible for Discount
- Product qualifies for Free Shipping
- Loan Approval

Using IF with AND

Example

=IF(AND(E2>=5,F2>=10000),

"Eligible",

"Not Eligible")

OR Function

Definition

The **OR** function returns TRUE if **at least one condition** is TRUE.

Only when all conditions are FALSE does it return FALSE.

Syntax

=OR(logical1,logical2,...)

Syntax Explanation

Argument	Meaning
logical1	First condition
logical2	Second condition
...	Additional conditions

Example

Quantity ≥ 5

OR

Price ≥ 10000

=OR(E2>=5,F2>=10000)

Returns

TRUE if any one condition is TRUE.

Truth Table

Condition 1	Condition 2	Result
TRUE	TRUE	TRUE
TRUE	FALSE	TRUE
FALSE	TRUE	TRUE
FALSE	FALSE	FALSE

Business Uses

- Promotional Offers
- Customer Eligibility
- Risk Detection
- Fraud Detection
- Approval Systems

Using IF with OR

=IF(OR(E2>=5,F2>=10000),

"Eligible",

"Not Eligible")

Difference Between AND and OR

AND	OR
All conditions must be TRUE	At least one condition must be TRUE
More restrictive	More flexible
Used for strict eligibility	Used for optional eligibility

Real Business Scenario

A retail company gives a **Premium Customer** tag if:

- Quantity ≥ 8 **AND**
- Price ≥ ₹20,000

Formula:

=IF(AND(E2>=8,F2>=20000),"Premium","Regular")

Another campaign offers a **Festival Discount** if:

- Customer purchased a Laptop **OR**
- Customer purchased a Smartphone

Formula:

=IF(OR(D2="Laptop",D2="Smartphone"),"Discount","No Discount")

Summary

- Use **IFS** to simplify multiple-condition decisions.
- Perform calculations with **SUMIFS**, **COUNTIFS**, and **AVERAGEIFS** using multiple criteria.
- Apply **AND** and **OR** to evaluate multiple logical conditions.
- Combine **IF** with **AND** and **OR** to solve real-world business problems.

- Build cleaner, more maintainable formulas that reflect common interview questions and business scenarios.